LAB5: CONFIGURATION OF STATIC ROUTES AND DEFAULT ROUTES

OBJECTIVE:

- To understand the concept of Static Routing and Default Routing for manual path selection in a network.
- To configure a network topology using Cisco Packet Tracer that connects different networks using Static and Default routes to visualise the flow of packets between different networks.

TOOLS USED

- Cisco Packet Tracer

THEORY:

Static Routing:

Static route implementation is the process of manually defining specific routing paths in a router's forwarding table. In this approach, network administrators directly configure the exact path that data packets must follow to reach a certain destination network.

Functionality:

The routes are configured using command-line instructions on the router. Whenever a packet arrives, the router checks these manually defined routes to determine where the packet should be forwarded. If the network path becomes unavailable or changes, administrators must manually update or delete the affected route.

```
ip route [destination_network] [subnet_mask] [next_hop_address]
```

Use:

Static routing is commonly applied for security purposes because it prevents routers from automatically sharing routing information. This helps hide certain network segments and reduces exposure to potential security threats.

Default Routing:

Default routing is a special form of static routing that functions as a "gateway of last resort." It directs the router to send packets to a particular next-hop router when the destination network is not present in the routing table.

This is configured by defining a route that uses an address made entirely of zeros. Such a

configuration instructs the router to forward any traffic with an unknown destination to a designated neighboring router.

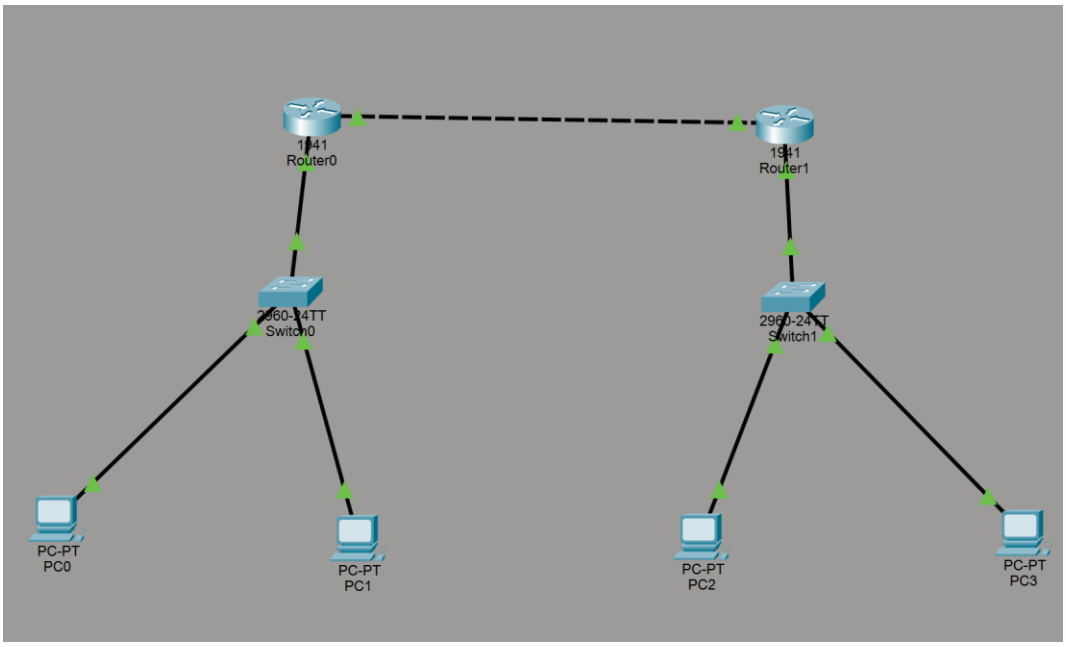
```
ip route 0.0.0.0 0.0.0.0 [next_hop_address]
```

Use Cases:

Default routing is typically implemented in **stub networks**, where there is only one path for outgoing traffic, or when a local network connects to the Internet through an Internet Service Provider (ISP). It also helps simplify routing by reducing the number of entries required in the routing table.

NETWORK TOPOLOGY:

The network infrastructure consists of two routing devices, interconnected through a serial communication link. Within the first segment, switch0 provides network access for workstations PC0 and PC1, connecting them to the device1. The second segment features Switch1, which enables connectivity for workstations PC2 and PC3 to device2. This configuration represents a basic multi-segment network requiring proper routing configuration for inter network communication.



Configuration Table :

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
Router 0	GigabitEthernet0/0	10.0.0.1	255.0.0.0	N/A
Rouer 0	GigabitEthernet0/1	192.168.1.1	255.255.255.0	N/A
Rouer 1	GigabitEthernet0/0	10.0.0.2	255.0.0.0	N/A
Rouer 1	GigabitEthernet0/1	192.168.2.1	255.255.255.0	N/A
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.1.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.1.3	255.255.255.0	192.168.2.1

Routing Configuration:

Device	Routing Type	Command/ Route
Router 0	Static Route	ip route 192.168.2.0 255.255.255.0 10.0.0.2
Rouer 1	Static Route	ip route 192.168.1.0 255.255.255.0 10.0.0.1
Rouer 0	Default Route(Alt)	ip route 0.0.0.0 0.0.0.0 10.0.0.2

Result:

The network was successfully configured. To verify the connectivity, a ping request was sent from PC1 (192.168.1.3) in the first network to PC2 (192.168.2.3) in the second network. The packets were sent and received successfully which validates that the static route is correct and similar was done for default route connection testing.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Discussion and Conclusion:

During this Lab we successfully implemented static and default routing using Cisco Packet Tracer. Static routing provided precise control over traffic paths and enhanced security, while default routing simplified the routing table for stub networks. The hands-on implementation demonstrated how routers forward packets based on implementation demonstrated how routers forward packets based on manually configured entries. The lab was completed successfully, providing essential knowledge of routing principles for network administration.